Basic Mechanics

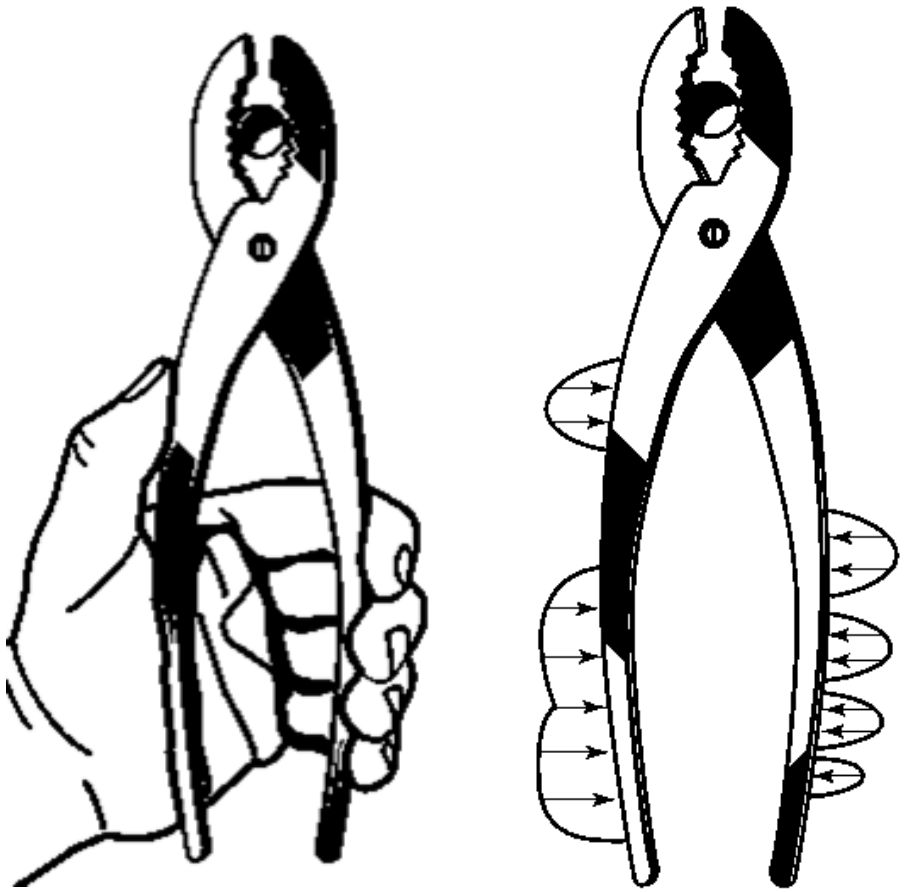
Equivalent Force Systems

ME 164/2

Department of Mechanical Engineering, KNUST

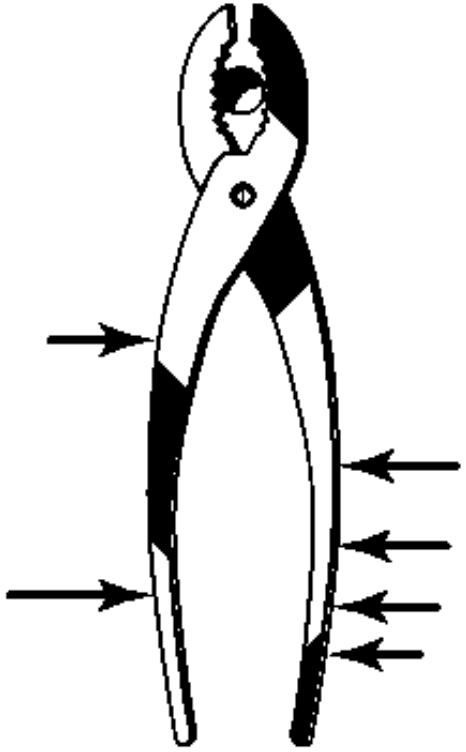
D. A. Quansah, PhD

Equivalent Force Systems

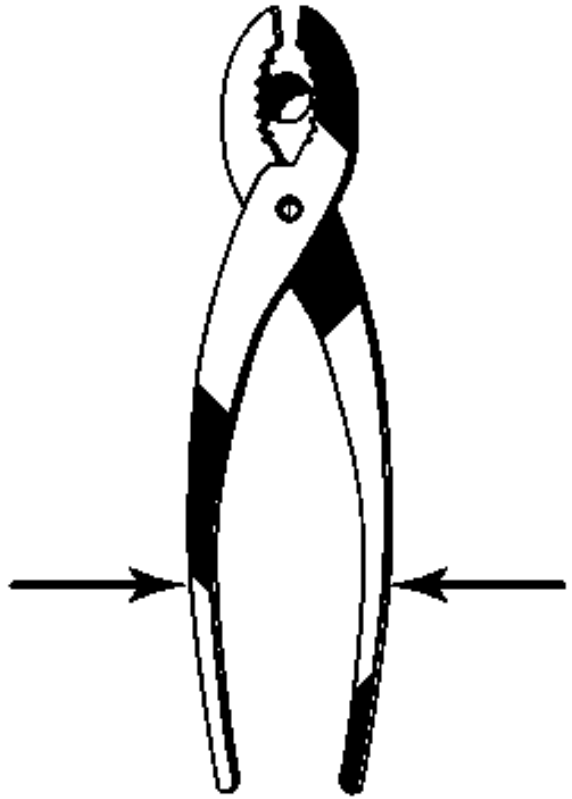


- For the pliers shown a force develops at the jaws as the handles are squeezed.
- Each finger and the palm of the hand apply a complex pressure distribution to the handles (as shown).
- However all complexities of these distributions need not be included for accurate engineering analyses.

Equivalent Force Systems



- In determining the force developed at the jaws, the pressure distributions may be replaced by **equivalent forces** with no loss of accuracy.



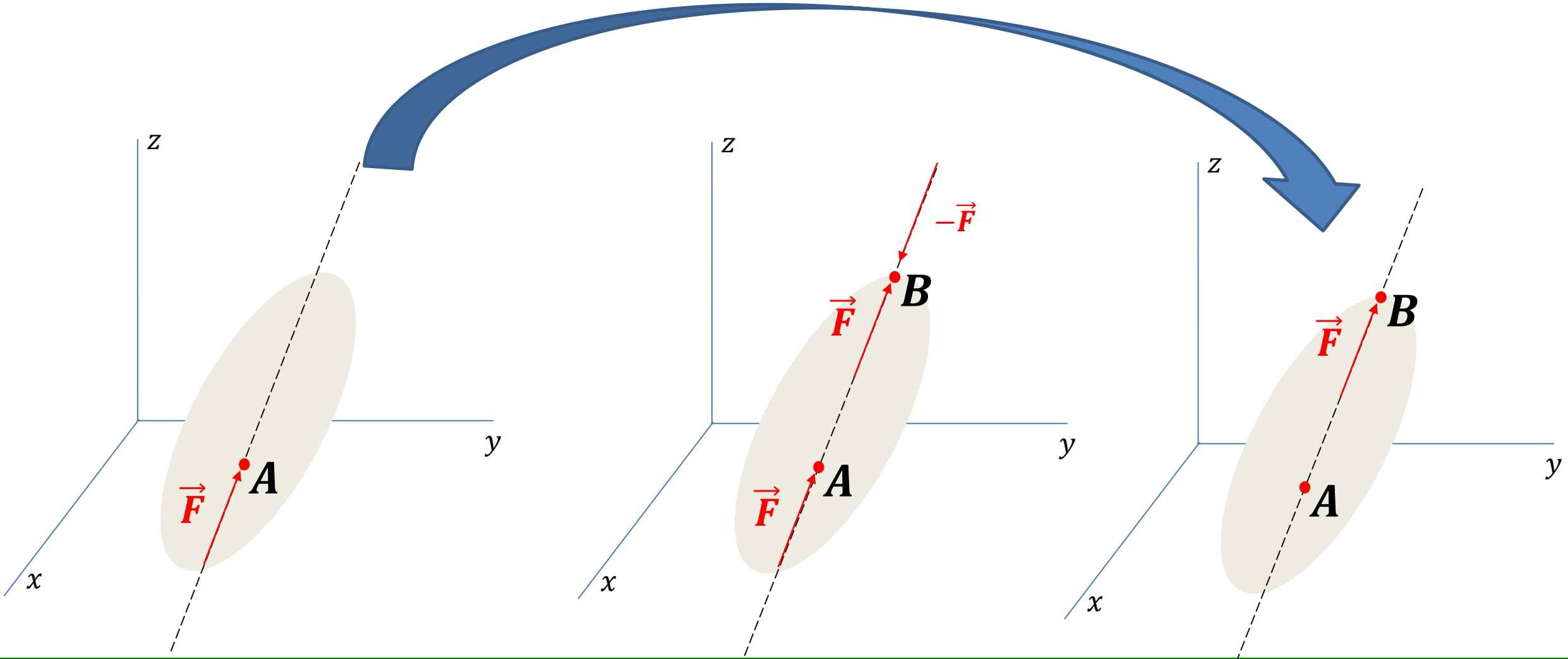
Transmissibility of a force

- A force applied to an object may produce external effects and internal effects.
- External effects refer to the response of the object as a whole.
- Examples of external effects include support reactions (for fixed bodies) or an object's displacement, velocity, and acceleration.
- Internal effects refer to the internal forces supported by a body, and if the body is deformable, the deformations that the body experiences.

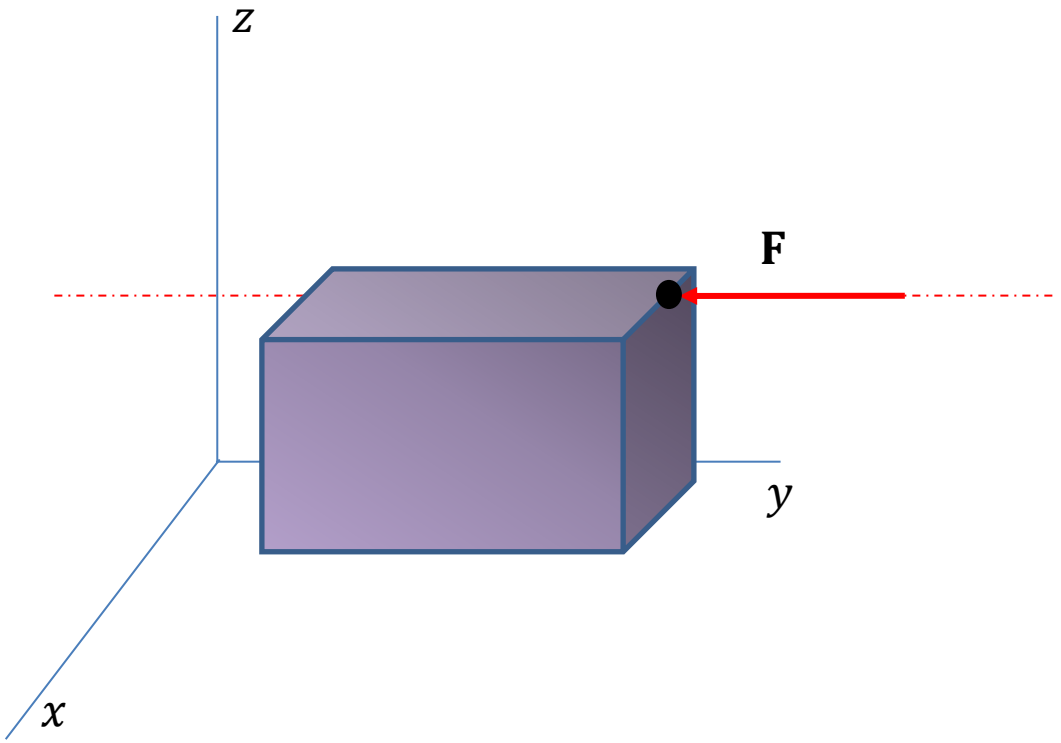
Transmissibility of a force

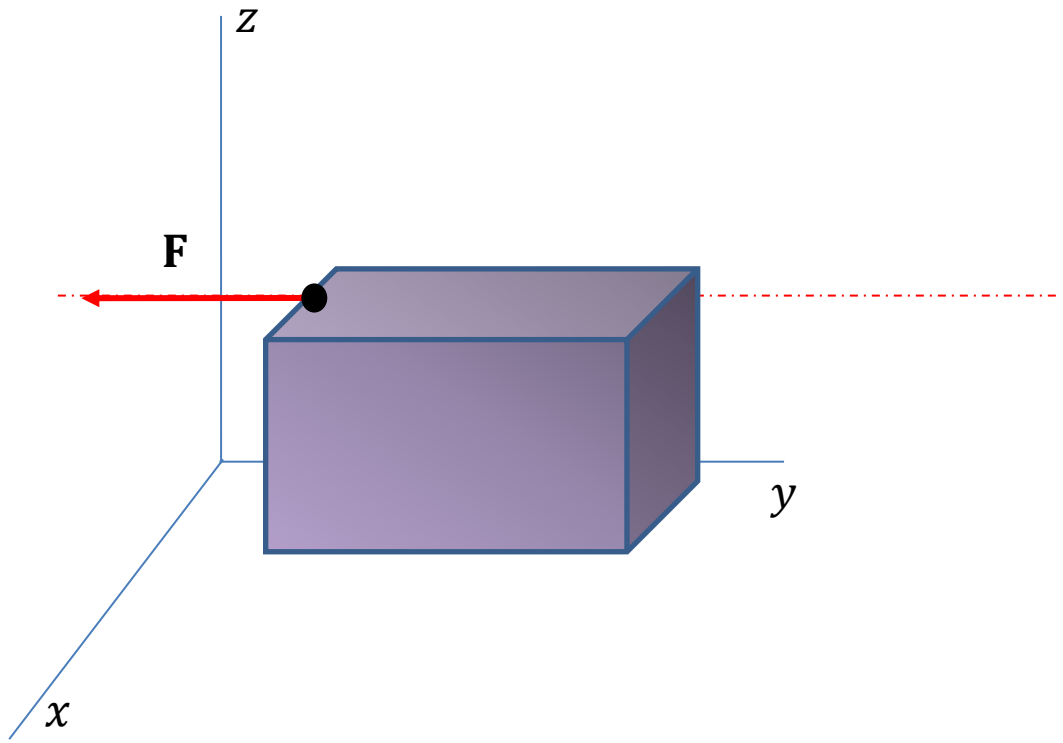
The principle of transmissibility states that the external effects of a force applied to a rigid body are the same, regardless of the point of application of the force along its line of action.

Transmissibility of a force



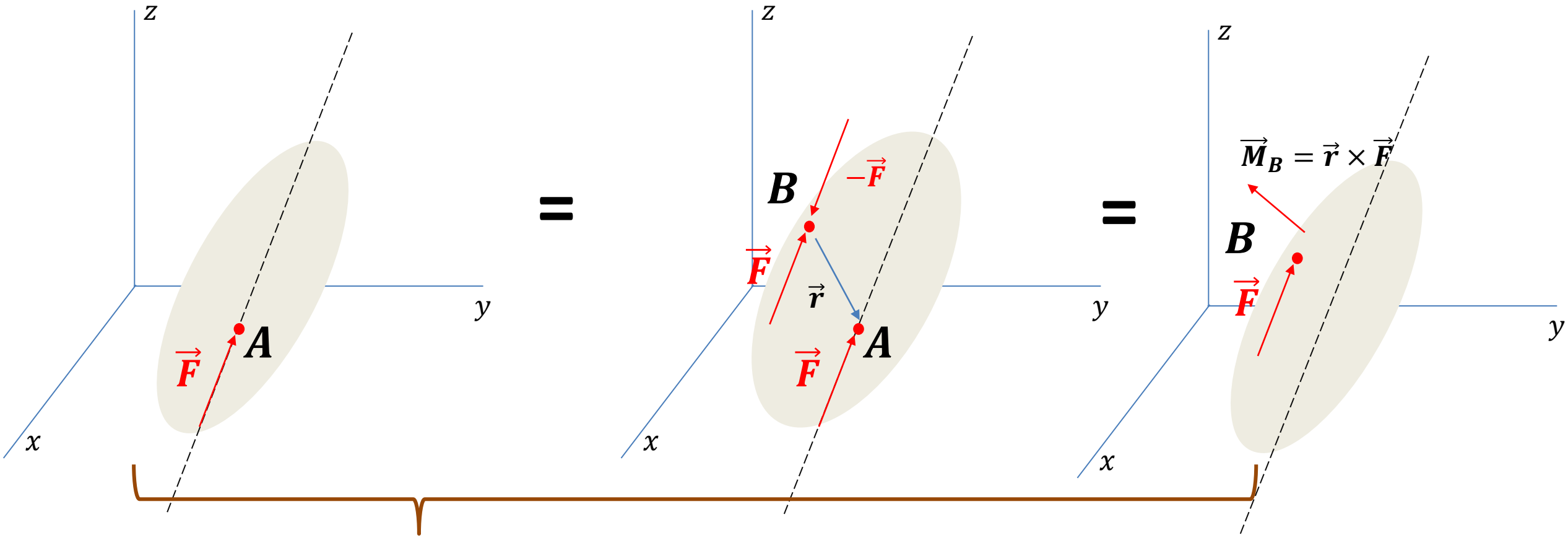
Transmissibility of a force





The principle of transmissibility suggests, that, the force F that will cause the block to slip, is the same regardless of where F is applied along its line of action.

Equivalent Force Systems



Two equivalent force systems

$\vec{F}$ and $-\vec{F}$ constitute a couple and may therefore be replaced by $\vec{M}_B = \vec{r} \times \vec{F}$

Equivalent Force Systems

- Two force systems are **equivalent** if they have the same **resultant force** and the same **resultant moment**.

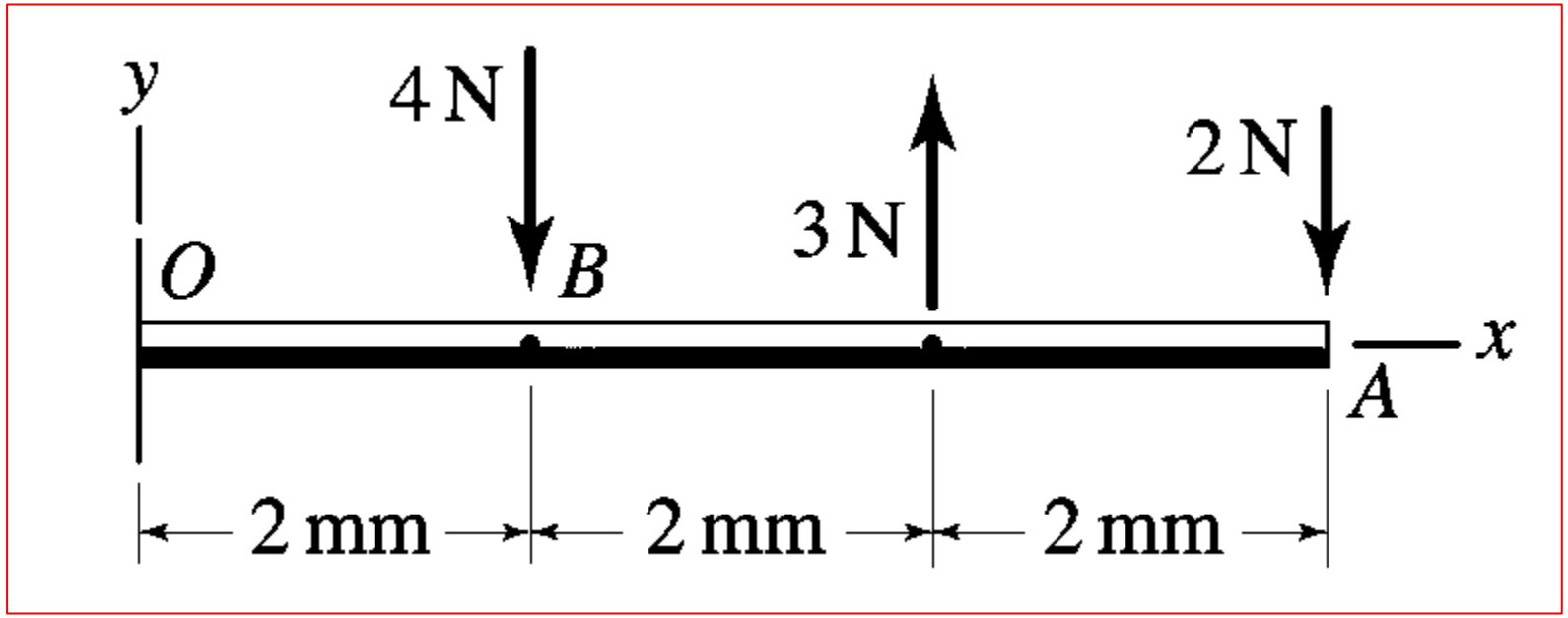
$$\left(\vec{F}_R\right)_{\text{system 1}} = \left(\vec{F}_R\right)_{\text{system 2}}$$

and

$$\left(\vec{M}_R\right)_{\text{system 1}} = \left(\vec{M}_R\right)_{\text{system 2}}$$

where the moment summation point is the same for system 1 and system 2.

Worked Example



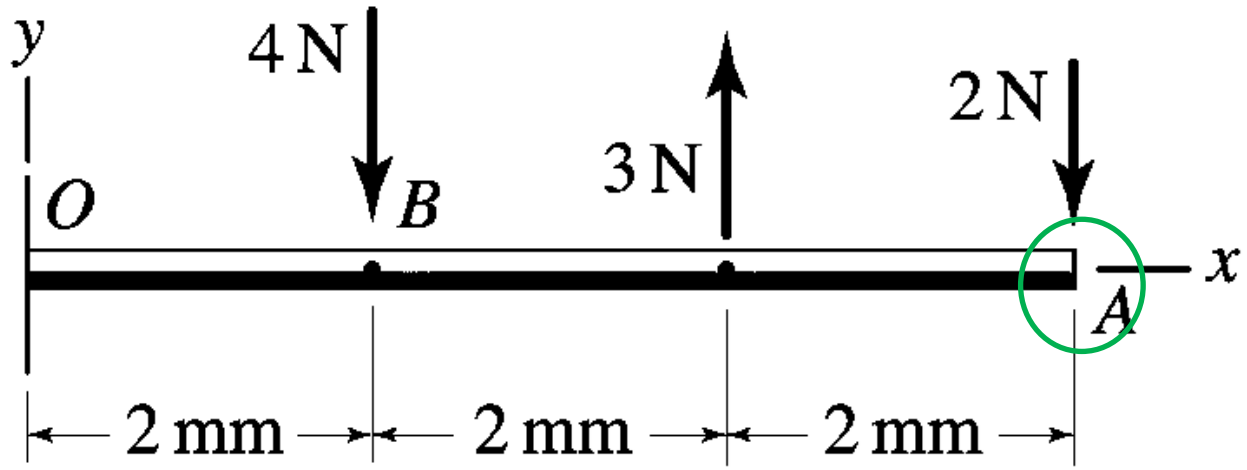
Determine an equivalent force system:

a) at point A,

b) at point B,

c) consisting of a single force only, and specify the x coordinate of the point where the force's line of action intersects the x axis.

Worked Example



Equivalent force system at point A:

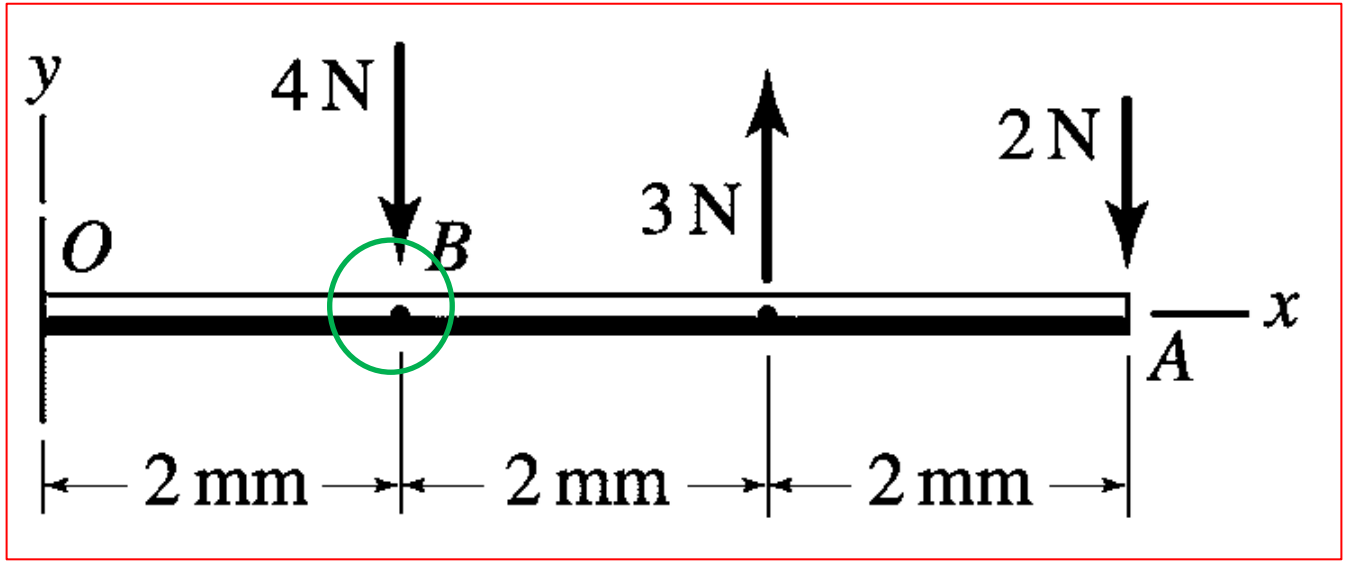
A force system applied at point A which produces the same external effect as what is shown in the diagram.

$$F_{Rx} = 0 \qquad F_{Ry} = -4\ \text{N} + 3\ \text{N} - 2\ \text{N} = -3\ \text{N}$$

$$M_{RA} = (4\ \text{N})(4\ \text{mm}) - (3\ \text{N})(2\ \text{mm})$$
$$= 10\ \text{N.mm}$$



Worked Example

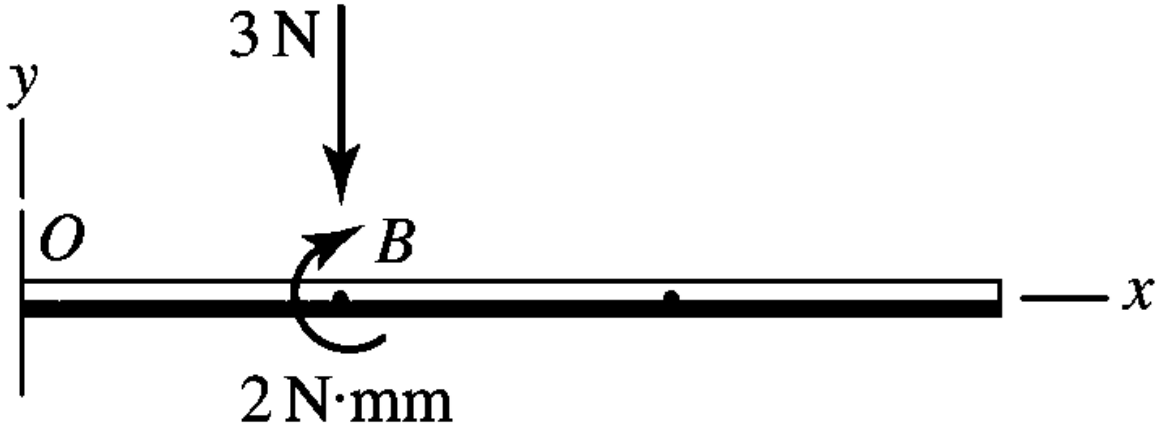


Equivalent force system at point B:

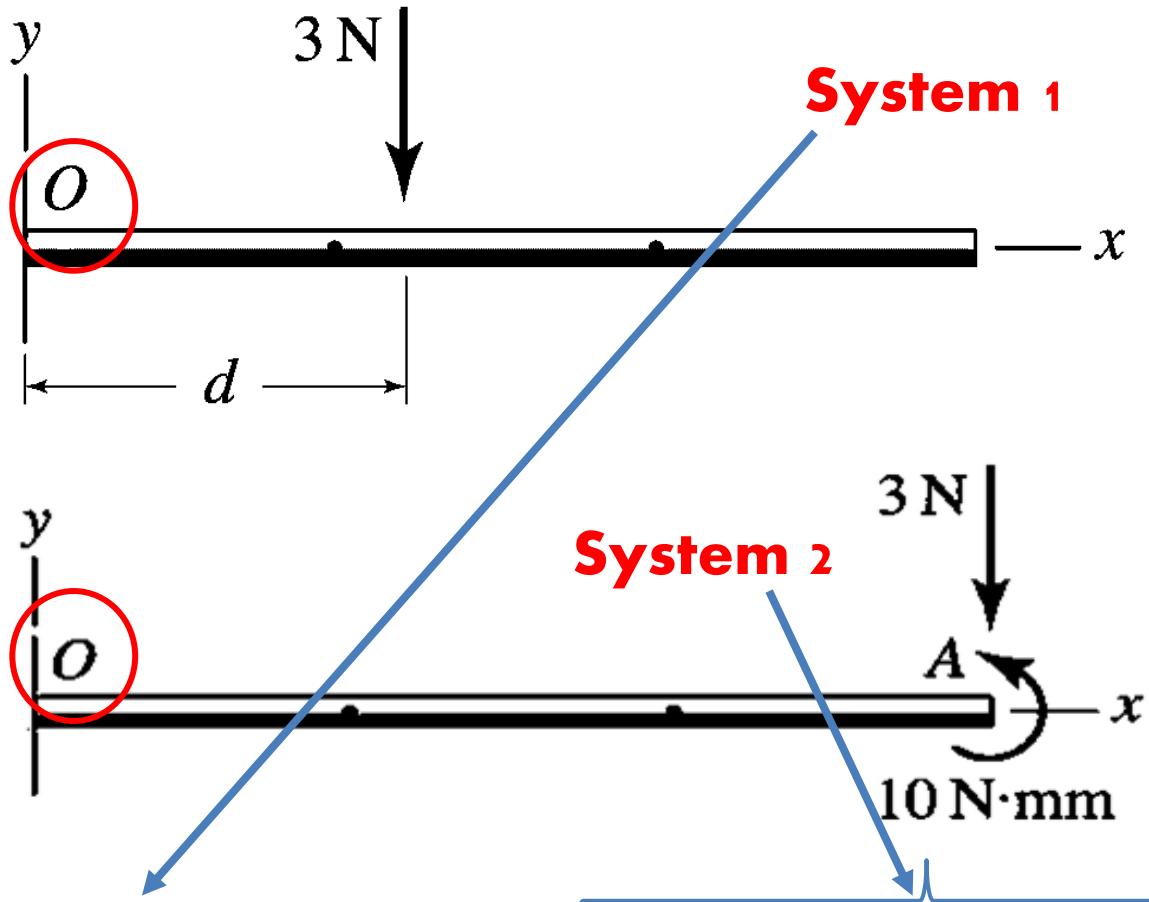
A force system applied at point B which produces the same external effect as what is shown in the diagram.

$$F_{Rx} = 0 \qquad F_{Ry} = -3\ \text{N}$$

$$M_{RB} = (3\ \text{N})(2\ \text{mm}) - (2\ \text{N})(4\ \text{mm})$$
$$= -2\ \text{N.mm}$$



Worked Example



$$-(3\ \mathrm{N})(d\ ) = -(3\ \mathrm{N})(6\ \mathrm{mm}) + 10\ \mathrm{N.mm}$$

$$d = 2.67\ .\mathrm{mm}$$

Equivalent force system at unknown point d:

A force system applied at point d which produces the same external effect as what is shown in the original diagram or its equivalents.

Since the resultant force is the same, the two systems must additionally, produce the same resultant moment about a given point - e.g. point O